

Martingale Detection of Treatment Effects in Wound Care

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Abstract

In clinical research, the ability to act on accumulating evidence motivates methods that support valid sequential decision-making. We apply a martingale-based framework for detecting treatment effects in a one-arm clinical trial in wound care, where outcomes are continuously compared against a reference distribution. We then extend this framework to individualized treatment evaluation by constructing patient-specific counterfactual trajectories from a real-world reference cohort using Gaussian kernel weighting. Because the underlying null distribution is not fully known, treatment effects are quantified through a martingale-inspired divergence measure, and statistical significance is assessed using an empirical null distribution generated from 5,000 resampled reference cohorts.

1. Introduction

The ability to detect treatment effects early is important in clinical trials, as ethical considerations may warrant stopping a study when clear benefit, harm, or futility becomes evident Pocock (2005). Assessing treatment effectiveness at the individual level is challenging because counterfactual outcomes are unobservable Rubin (1974). In wound care, patient characteristics beyond the wound itself may strongly influence healing trajectories Almadani et al. (2021), making population-level treatment effects difficult to translate to individual patients. We consider a martingale-based approach for sequential treatment-effect evaluation, where observed outcomes are compared against a reference distribution and evidence is accumulated over time. Further, we extend this idea by constructing patient-specific reference trajectories from a real-world cohort using similarity-based weighting.

2. Method

In the classical test-martingale setting, a reference distribution F_0 is assumed known. Letting Y_t denote the observed outcome and $\mu_t = \mathbb{E}_{F_0}[Y_t]$ the expected outcome under F_0 , evidence is accumulated through the residuals $\epsilon_t = Y_t - \mu_t$ Shafer et al. (2011). In practice, however, the counterfactual outcome distribution is rarely known. We therefore, in addition to the traditional approach, estimate an individualized reference distribution from a real-world wound care registry. Let $(X_i, Y_i)_{i=1}^n$ denote patients in a reference cohort (RC) representing outcomes under high-quality wound care from the Wound Monitor database Gustafsson et al. (2026). For a target patient t with covariates X_t , we construct a personalized reference outcome

$$w_i(X_t) = \exp\left(-\frac{\|X_i - X_t\|^2}{2h^2}\right), \quad \mu(X_t) = \frac{\sum_{i=1}^n w_i(X_t)Y_i}{\sum_{i=1}^n w_i(X_t)}.$$

Using the residual $\epsilon_t = Y_t - \mu(X_t)$, we define

$$M_0 = 1, \quad M_t = \max(10^{-6}, M_{t-1}(1 + \lambda \tilde{\epsilon}_t)), \quad t \geq 1.$$

where λ controls the bet size and $\tilde{\epsilon}_t = z_t \mathbf{1}(|z_t| > \gamma)$ retains only large standardized deviations. In the idealized setting where the reference distribution is known, (M_t) forms a test martingale under the null hypothesis. In the setting where the reference distribution is estimated from registry data, we do not make this assumption. Instead, we assess significance empirically by generating 5,000 pseudo-target cohorts from the RC and comparing their martingale growth to that observed in the target cohort. For each iteration, the patients not assigned to the pseudo-target cohort constitute the RC. The resulting p-value is the proportion of pseudo-cohorts whose martingale exceeds that of the target cohort.

3. Results

We omit results with the traditional approach due to space limitations, it will be included in the poster. The RC martingale attained a maximum value of $\sup(M_{80}) = 29.02$, corresponding to a p-value of 0.034 under the assumption that M has true martingale properties which is true if $E[\mu(X_t)] = E[Y_t]$. In contrast, the pairwise comparison found that the TC martingale exceeded the maximum value of the sampled trajectories in 87.7% of runs, corresponding to a p-value of 0.123. Figure 1 summarizes both analyses.

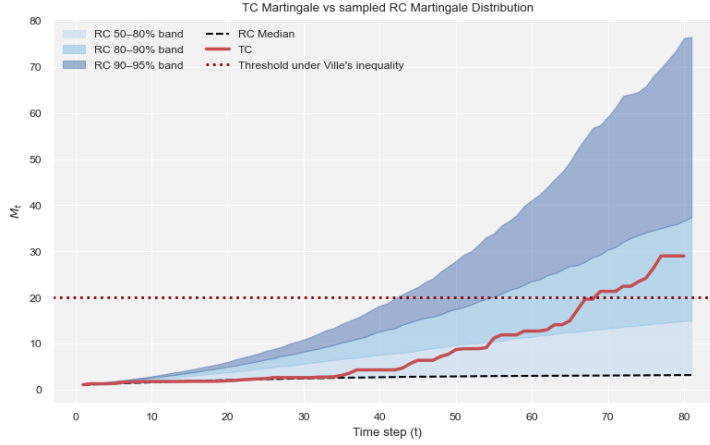


Figure 1: Comparison of TC Martingale to 5000 subsampled martingales. For vizualation purposes, a single run of TC compared to the full RC is shown.

4. Conclusion

This work extends test-martingale methodology to settings where individualized counterfactual trajectories must be estimated from real-world data. The results illustrate the importance of not assuming the martingale property unless it is theoretically justified.

Future work includes refining kernel selection, improving robustness, extending the method to time-varying covariates as well as investigating alternative betting functions.

References

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